

Core Signatures and Inversions

Supplementary material: substitution rules, closed-form solutions and proofs

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This supplement accompanies the manuscript *Core Signatures and Inversions* (Wilmott, WP-1259, revision 1). Section S1 tabulates the 48 substitution rules through level five for the binary alphabet; Section S2 gives the closed-form solutions of the two-segment Chen system and the three-segment matching system; Section S3 proves Proposition 1 (core variables are Lyndon coordinates) and Section S4 proves the three stability theorems. References to the main text are marked as such; the theorems, lemmas and equations numbered here belong to this supplement.

S1. Substitution rules Q_m for $d = 2$

The substitution rules express every non-core signature entry as a polynomial in core entries. The cores referenced below are

$$C_1 = \{s_1, s_2\}, \quad C_2 = C_1 \cup \{s_{1,2}\}, \quad C_3 = C_2 \cup \{s_{1,1,2}, s_{1,2,2}\},$$

$$C_4 = C_3 \cup \{s_{1,1,1,2}, s_{1,1,2,2}, s_{1,2,2,2}\},$$

$$C_5 = C_4 \cup \{s_{1,1,1,1,2}, s_{1,1,1,2,2}, s_{1,1,2,1,2}, s_{1,1,2,2,2}, s_{1,2,1,2,2}, s_{1,2,2,2,2}\}.$$

Within each primitive rotation class of words the representative kept is the lexicographically smallest rotation, the Lyndon word of the class (Section S3); a letter multiset can contain several Lyndon words (11122 and 11212 at length five).

Level 2. $s_{1,1} = \frac{1}{2}s_1^2, \quad s_{2,1} = s_1s_2 - s_{1,2}, \quad s_{2,2} = \frac{1}{2}s_2^2.$

Level 3. $s_{1,1,1} = \frac{1}{6}s_1^3, \quad s_{1,2,1} = s_1s_{1,2} - 2s_{1,1,2}, \quad s_{2,1,1} = -s_1s_{1,2} + s_{1,1,2} + \frac{1}{2}s_2s_1^2, \quad s_{2,1,2} = s_2s_{1,2} - 2s_{1,2,2}, \quad s_{2,2,1} = -s_2s_{1,2} + s_{1,2,2} + \frac{1}{2}s_1s_2^2, \quad s_{2,2,2} = \frac{1}{6}s_2^3.$

Level 4. $s_{1,1,1,1} = s_1^4/24, \quad s_{1,1,2,1} = s_1s_{1,1,2} - 3s_{1,1,1,2}, \quad s_{1,2,1,1} = s_1^2s_{1,2}/2 - 2s_1s_{1,1,2} + 3s_{1,1,1,2}, \quad s_{1,2,1,2} = -2s_{1,1,2,2} + s_{1,2}^2/2, \quad s_{1,2,2,1} = s_1s_{1,2,2} - s_{1,2}^2/2, \quad s_{2,1,1,1} = s_1^3s_2/6 - s_1^2s_{1,2}/2 + s_1s_{1,1,2} - s_{1,1,1,2}, \quad s_{2,1,1,2} = s_{1,1,2}s_2 - s_{1,2}^2/2, \quad s_{2,1,2,1} = s_1s_{1,2}s_2 - 2s_{1,1,2,2} - 2s_{1,1,2}s_2 + 2s_{1,1,2,2} + s_{1,2}^2/2, \quad s_{2,1,2,2} = s_{1,2,2}s_2 - 3s_{1,2,2,2}, \quad s_{2,2,1,1} = s_1^2s_2^2/4 - s_1s_{1,2}s_2 + s_1s_{1,2,2} + s_{1,1,2}s_2 - s_{1,1,2,2}, \quad s_{2,2,1,2} = s_{1,2}s_2^2/2 - 2s_{1,2,2,2} + 3s_{1,2,2,2}, \quad s_{2,2,2,1} = s_1s_2^3/6 - s_{1,2}s_2^2/2 + s_{1,2,2}s_2 - s_{1,2,2,2}, \quad s_{2,2,2,2} = s_2^4/24.$

Level 5. $s_{1,1,1,1,1} = s_1^5/120, \quad s_{1,1,1,2,1} = s_1s_{1,1,1,2} - 4s_{1,1,1,1,2}, \quad s_{1,1,2,1,1} = s_1^2s_{1,1,2}/2 - 3s_1s_{1,1,1,2} + 6s_{1,1,1,1,2}, \quad s_{1,1,2,2,1} = s_1s_{1,1,2,2} - 3s_{1,1,1,2,2} - s_{1,1,2,1,2}, \quad s_{1,2,1,1,1} = s_1^3s_{1,2}/6 - s_1^2s_{1,1,2} + 3s_1s_{1,1,1,2} - 4s_{1,1,1,1,2}, \quad s_{1,2,1,1,2} = -6s_{1,1,1,2,2} + s_{1,1,2}s_{1,2} - 3s_{1,1,2,1,2}, \quad s_{1,2,1,2,1} = -2s_1s_{1,1,2,2} + s_1s_{1,2}^2/2 + 12s_{1,1,1,2,2} - 2s_{1,1,2}s_{1,2} + 4s_{1,1,2,1,2}, \quad s_{1,2,2,1,1} = s_1^2s_{1,2,2}/2 - s_1s_{1,2}^2/2 - 3s_{1,1,1,2,2} + s_{1,1,2}s_{1,2} - s_{1,1,2,1,2}, \quad s_{1,2,2,1,2} = -6s_{1,1,2,2,2} + s_{1,2}s_{1,2,2} - 3s_{1,2,1,2,2}, \quad s_{1,2,2,2,1} = s_1s_{1,2,2,2} + 4s_{1,1,2,2,2} - s_{1,2}s_{1,2,2} + 2s_{1,2,1,2,2}, \quad s_{2,1,1,1,1} = s_1^4s_2/24 - s_1^3s_{1,2}/6 + s_1^2s_{1,1,2}/2 - s_1s_{1,1,1,2} + s_{1,1,1,1,2}, \quad s_{2,1,1,1,2} = s_{1,1,1,2}s_2 + 4s_{1,1,1,2,2} - s_{1,1,2}s_{1,2} + 2s_{1,1,2,1,2}, \quad s_{2,1,1,2,1} = s_1s_{1,1,2}s_2 - s_1s_{1,2}^2/2 - 3s_{1,1,1,2,2} - 6s_{1,1,1,2,2} + 2s_{1,1,2}s_{1,2} - 3s_{1,1,2,1,2}, \quad s_{2,1,1,2,2} = s_{1,1,2,2}s_2 - 3s_{1,1,2,2,2} - s_{1,2,1,2,2}, \quad s_{2,1,2,1,1} = s_1^2s_{1,2}s_2/2 - s_1^2s_{1,2,2} - 2s_1s_{1,1,2,2} + 2s_1s_{1,1,2,2} +$

$$\begin{aligned}
& s_1 s_{1,2}^2/2 + 3s_{1,1,1,2}s_2 - s_{1,1,2}s_{1,2} + s_{1,1,2,1,2}, \quad s_{2,1,2,1,2} = -2s_{1,1,2,2}s_2 + 12s_{1,1,2,2,2} + s_{1,2}^2 s_2/2 - \\
& 2s_{1,2}s_{1,2,2} + 4s_{1,2,1,2,2}, \quad s_{2,1,2,2,1} = s_1 s_{1,2,2}s_2 - 3s_1 s_{1,2,2,2} - 6s_{1,1,2,2,2} - s_{1,2}^2 s_2/2 + 2s_{1,2}s_{1,2,2} - 3s_{1,2,1,2,2}, \\
& s_{2,1,2,2,2} = s_{1,2,2,2}s_2 - 4s_{1,2,2,2,2}, \quad s_{2,2,1,1,1} = s_1^3 s_2^2/12 - s_1^2 s_{1,2}s_2/2 + s_1^2 s_{1,2,2}/2 + s_1 s_{1,1,2}s_2 - \\
& s_1 s_{1,1,2,2} - s_{1,1,1,2}s_2 + s_{1,1,1,2,2}, \quad s_{2,2,1,1,2} = s_{1,1,2}s_2^2/2 - 3s_{1,1,2,2,2} - s_{1,2}^2 s_2/2 + s_{1,2}s_{1,2,2} - s_{1,2,1,2,2}, \\
& s_{2,2,1,2,1} = s_1 s_{1,2}s_2^2/2 - 2s_1 s_{1,2,2}s_2 + 3s_1 s_{1,2,2,2} - s_{1,1,2}s_2^2 + 2s_{1,1,2,2}s_2 + s_{1,2}^2 s_2/2 - s_{1,2}s_{1,2,2} + s_{1,2,1,2,2}, \\
& s_{2,2,1,2,2} = s_{1,2,2}s_2^2/2 - 3s_{1,2,2,2}s_2 + 6s_{1,2,2,2,2}, \quad s_{2,2,2,1,1} = s_1^2 s_2^3/12 - s_1 s_{1,2}s_2^2/2 + s_1 s_{1,2,2}s_2 - \\
& s_1 s_{1,2,2,2} + s_{1,1,2}s_2^2/2 - s_{1,1,2,2}s_2 + s_{1,1,2,2,2}, \quad s_{2,2,2,1,2} = s_{1,2}s_2^3/6 - s_{1,2,2}s_2^2 + 3s_{1,2,2,2}s_2 - 4s_{1,2,2,2,2}, \\
& s_{2,2,2,2,1} = s_1 s_2^4/24 - s_{1,2}s_2^3/6 + s_{1,2,2}s_2^2/2 - s_{1,2,2,2}s_2 + s_{1,2,2,2,2}, \quad s_{2,2,2,2,2} = s_2^5/120.
\end{aligned}$$

The rules were generated by the shuffle-product elimination of Section 3 of the main text and checked against the signatures of random piecewise-linear paths computed directly from Chen's identity; the largest discrepancy over the 48 rules is 4×10^{-16} . Each rule has the form that Proposition 1 of the main text predicts: the word's Lyndon factorisation supplies the leading product of core entries, and every other term is a product of core entries of lexicographically smaller words.

S2. Closed-form solutions of the time-ordered Chen systems

Two pieces. The system of Table 3 of the main text has the unknowns T, t_1, β_1, β_2 and the five equations at levels 1–3. Dropping equation #5 ($z_{1,2,2}$) and solving the remaining four gives Solution 1:

$$\begin{aligned}
T &= z_1, \quad t_1 = \frac{2(z_1 z_{1,2} - 3z_{1,1,2})}{z_1 z_2 - 2z_{1,2}}, \\
\beta_1 &= \frac{z_1^2 z_2^2 - 2z_1 z_2 z_{1,2} + 4z_{1,2}^2 - 6z_2 z_{1,1,2}}{2z_1(z_1 z_{1,2} - 3z_{1,1,2})}, \quad \beta_2 = \frac{2(3z_2 z_{1,1,2} - 2z_{1,2}^2)}{z_1(z_1^2 z_2 - 4z_1 z_{1,2} + 6z_{1,1,2})}. \quad (\text{S2.1})
\end{aligned}$$

Dropping equation #4 ($z_{1,1,2}$) instead gives Solution 2:

$$\begin{aligned}
T &= z_1, \quad t_1 = \frac{2(3z_1 z_{1,2,2} - 2z_{1,2}^2)}{z_2(z_1 z_2 - 2z_{1,2})}, \\
\beta_1 &= \frac{z_2(z_1 z_2^2 - 4z_2 z_{1,2} + 6z_{1,2,2})}{2(3z_1 z_{1,2,2} - 2z_{1,2}^2)}, \quad \beta_2 = \frac{2z_2(z_2 z_{1,2} - 3z_{1,2,2})}{z_1^2 z_2^2 - 2z_1 z_2 z_{1,2} + 4z_{1,2}^2 - 6z_1 z_{1,2,2}}. \quad (\text{S2.2})
\end{aligned}$$

The middle point $P_1 = (t_1, \beta_1 t_1)$ of each solution is

$$P_1^{(1)} = \left(\frac{2(z_1 z_{1,2} - 3z_{1,1,2})}{z_1 z_2 - 2z_{1,2}}, \frac{z_1^2 z_2^2 - 2z_1 z_2 z_{1,2} + 4z_{1,2}^2 - 6z_2 z_{1,1,2}}{z_1(z_1 z_2 - 2z_{1,2})} \right), \quad (\text{S2.3})$$

$$P_1^{(2)} = \left(\frac{2(3z_1 z_{1,2,2} - 2z_{1,2}^2)}{z_2(z_1 z_2 - 2z_{1,2})}, \frac{z_1 z_2^2 - 4z_2 z_{1,2} + 6z_{1,2,2}}{z_1 z_2 - 2z_{1,2}} \right). \quad (\text{S2.4})$$

Substituting Solution 1 into equation #5 leaves the residual $(z_1 z_2 z_{1,2} + 2z_{1,2}^2 - 3z_1 z_{1,2,2} - 3z_2 z_{1,1,2})/(3z_1)$, and Solution 2 into equation #4 the same numerator over $3z_2$: where the respective formulas are defined, the residuals vanish exactly when the consistency condition holds. If both formulas are defined, the solutions then coincide. Admissibility still requires $0 < t_1 < T$. The denominators show the degenerate cases discussed there: $z_1 z_2 - 2z_{1,2} = 0$ (the Lévy area relative to the chord is zero, so the path is indistinguishable at level 2 from its chord) and, for Solution 2, $z_2 = 0$. The solutions were obtained by elimination in a computer algebra system and checked by substitution into the retained equations and by calculating the displayed omitted-equation residuals.

Three pieces. With slopes $\beta_1, \beta_2, \beta_3$ on $[0, t_1]$, $[t_1, t_2]$, $[t_2, T]$ and X_i the i -th segment, Chen's identity gives every entry of the concatenation as

$$\sigma_w(X_1 * X_2 * X_3) = \sum_{w=uvy} \sigma_u(X_1) \sigma_v(X_2) \sigma_y(X_3), \quad \sigma_u(X_i) = \frac{\Delta t_i^{|u|_1} (\beta_i \Delta t_i)^{|u|_2}}{|u|!}, \quad (\text{S2.5})$$

the sum over all splittings of w into three (possibly empty) consecutive factors, $|u|_a$ the number of letters a in u , and $\Delta t_1 = t_1$, $\Delta t_2 = t_2 - t_1$, $\Delta t_3 = T - t_2$. The three-piece system equates (S2.5) to the target for the eight core words of length ≤ 4 : six unknowns $T, t_1, t_2, \beta_1, \beta_2, \beta_3$, eight equations. The single-letter-2 equations are the moment conditions of Proposition 5 of the main text:

$$T = z_1, \quad \sum_{i=1}^3 \beta_i \frac{t_i^{k+1} - t_{i-1}^{k+1}}{(k+1)!} = z_{1^k 2}, \quad k = 0, 1, 2, 3,$$

with $t_0 = 0$, $t_3 = T$; the three remaining equations, for $z_{1,2,2}$, $z_{1,1,2,2}$ and $z_{1,2,2,2}$, are polynomials of degrees 2, 2 and 3 in the slopes obtained by expanding (S2.5) (the last has forty monomials and is not printed). Selecting six equations gives a formally square subsystem, whose solvability and independence require checking; omitted equations still have to be satisfied for exact recovery; in practice we solve the three-piece and larger systems by the regression of Section 4.3 of the main text, which uses all the equations at once.

S3. Proof of Proposition 1 of the main text

The proof assembles four classical results — the Lyndon basis theorem, Poincaré–Birkhoff–Witt, Radford's theorem and Lyndon factorisation — and one computation, the elimination of Section 3 of the main text. We state the results in the form used and give references for their proofs; the two lemmas and the proof of the proposition are ours.

Setting. Let $V = \mathbb{R}^d$ with basis $\mathcal{A} = \{1, \dots, d\}$, let $T(V) = \bigoplus_{n \geq 0} V^{\otimes n}$ be the tensor algebra with the words on $\mathcal{A}$ as basis, and $T((V))$ its completion, where signatures live. The vector space $T(V)$ carries two Hopf algebra structures ([Reutenauer (1993)] Sections 1.4–1.6): the *concatenation–desshuffle* structure $T(V)_\bullet$, whose product is juxtaposition of words and whose coproduct is defined on letters by $\Delta_\bullet(a) = a \otimes 1 + 1 \otimes a$ and extended multiplicatively; and the *shuffle–deconcatenation* structure $T(V)_\sqcup$, whose product is the shuffle $\sqcup$ of (4) and whose coproduct is $\Delta_\bullet(w) = \sum_{w=uv} u \otimes v$. The two are graded-dual under the pairing $\langle u, v \rangle = \delta_{u,v}$ on words.

A series $g \in T((V))$ with empty-word coefficient 1 is *group-like* if $\Delta_\sqcup(g) = g \otimes g$. The signature of a bounded-variation path is group-like: writing $\sigma_w = \langle S(X), w \rangle$, the shuffle identity (4), $\sigma_I \sigma_J = \sum_{K \in I \sqcup J} \sigma_K$, says exactly that the linear functional $w \mapsto \sigma_w$ is an algebra homomorphism $(T(V), \sqcup) \rightarrow \mathbb{R}$, that is, a *character* of the shuffle algebra; and characters of $T(V)_\sqcup$ are the group-like elements of $T((V))_\bullet$. The set of truncated signatures

$$G^m(\mathbb{R}^d) = \{S^{(\leq m)}(X) : X \text{ of bounded variation in } \mathbb{R}^d\} \subset T^m(V)$$

is the step- m free nilpotent Lie group on d generators, an algebraic variety cut out by the shuffle relations $\sigma_I \sigma_J = \sum_{K \in I \sqcup J} \sigma_K$ with $|I| + |J| \leq m$ (total degree meaning word length); [Amendola, Friz and Sturmfels (2019)] call it the universal variety and identify its coordinate ring with the polynomial algebra in the entries σ_w , $|w| \leq m$, modulo the truncated shuffle ideal $\mathcal{I}_\sqcup^{(\leq m)}$.

The four classical results.

Theorem 1 (Lyndon basis; [Reutenauer (1993)] Theorem 5.1). *Define the bracket of a Lyndon word recursively by $[a] = a$ for a letter and $[w] = [[u], [v]]$ for the standard factorisation $w = uv$ with v the longest proper Lyndon suffix of w . The brackets $\{[w] : w \text{ Lyndon on } \mathcal{A}\}$ form a homogeneous basis of the free Lie algebra $\mathfrak{L}(V)$. In particular the number of Lyndon words of length n equals $\dim \mathfrak{L}^n(V)$, which is Witt's formula (7).*

Theorem 2 (Poincaré–Birkhoff–Witt; [Reutenauer (1993)] Theorem 0.5). *The universal enveloping algebra of $\mathfrak{L}(V)$ is $T(V)_\bullet$, with $\mathfrak{L}(V) \hookrightarrow T(V)_\bullet$ given by $[a, b] = a \otimes b - b \otimes a$ on letters. Consequently the ordered monomials $[w_1]^{\otimes n_1} \otimes \cdots \otimes [w_k]^{\otimes n_k}$, $w_1 < \cdots < w_k$ Lyndon, $n_i \geq 1$, form a basis of $T(V)_\bullet$.*

Theorem 3 (Radford; [Radford (1979)], [Reutenauer (1993)] Theorem 6.1). *The shuffle algebra $(T(V), \sqcup)$ is the free commutative algebra on the set $\mathcal{L}$ of Lyndon words: every element of $T(V)$ is uniquely a polynomial in Lyndon words under the shuffle product. Moreover, if $w = l_1 l_2 \cdots l_k$ with $l_1 \geq \cdots \geq l_k$ Lyndon is the Lyndon factorisation of w , and $n_1, \dots, n_r$ are the multiplicities of the distinct factors, then*

$$l_1 \sqcup l_2 \sqcup \cdots \sqcup l_k = n_1! \cdots n_r! w + \sum_{v < w} c_v v, \quad c_v \in \mathbb{Z}_{\geq 0}, \quad (\text{S3.1})$$

the sum running over words v lexicographically smaller than w with the same letter multiset.

Lemma 4 (Lyndon factorisation; [Reutenauer (1993)], Chapter 5). *Every word w on $\mathcal{A}$ has a unique factorisation $w = l_1 l_2 \cdots l_k$ into Lyndon words with $l_1 \geq \cdots \geq l_k$. A primitive word (one that is not a proper power) has exactly one Lyndon word among its rotations, namely the lexicographically smallest rotation.*

Two lemmas.

Lemma 5 (Lyndon entries are a transcendence basis). *For each $m \geq 1$,*

$$\mathcal{O}(G^m(\mathbb{R}^d)) = \mathbb{R}[\sigma_w : |w| \leq m] / \mathcal{I}_\sqcup^{(\leq m)} \cong \mathbb{R}[\sigma_w : w \in \mathcal{L}, |w| \leq m], \quad (\text{S3.2})$$

and every σ_v with v non-Lyndon, $|v| \leq m$, equals a unique polynomial P_v in the entries σ_w , w Lyndon, $|w| \leq |v|$.

Proof. Evaluate a character χ of $(T(V), \sqcup)$ on (S3.1): since χ is multiplicative for $\sqcup$,

$$\chi(l_1) \cdots \chi(l_k) = n_1! \cdots n_r! \chi(w) + \sum_{v < w} c_v \chi(v).$$

Induct on the lexicographic order within each length: the smallest word of length n is 1^n , whose only factor is the letter 1 with multiplicity n , so $\chi(1^n) = \chi(1)^n/n!$ and no sum appears. For a general non-Lyndon w , every $v < w$ in the sum is either Lyndon or has already been expressed, so $\chi(w)$ is a polynomial in the Lyndon values of length $\leq |w|$; this is P_w , and its existence for every character means $\sigma_w - P_w(\sigma)$ vanishes on G^m . Uniqueness: two such polynomials would give a polynomial relation among the Lyndon entries valid on every group-like element, and by Theorem 3 the Lyndon words are free generators of $(T(V), \sqcup)$, so the characters take arbitrary values on them (a character is determined by, and may be prescribed freely on, the free generators) and the relation is trivial.

For (S3.2), the polynomials P_v for $|v| \leq m$ are derived from shuffle relations of total word weight $\leq m$, so modulo $\mathcal{I}_\sqcup^{(\leq m)}$ every entry is a polynomial in the Lyndon entries of length $\leq m$, and the quotient is generated by them; by the uniqueness just shown they satisfy no polynomial relation on G^m , so the quotient is the polynomial ring on them. The vanishing ideal of G^m contains $\mathcal{I}_\sqcup^{(\leq m)}$ and the quotient by the latter already has no further relation valid on G^m , so the two ideals coincide and the quotient is the coordinate ring. (That the truncated signatures of bounded-variation paths fill the whole group G^m , so that “valid on G^m ” and “valid on every group-like element of $T^m(V)$ ” agree, is Chow’s theorem; piecewise-linear paths suffice.) $\square$

Lemma 6 (The elimination keeps the Lyndon representatives). *Fix a level n and process the words of length n in increasing lexicographic order. Solve σ_w whenever a relation generated by the shuffle identities (4) expresses it as a polynomial in entries of lexicographically smaller words of length n and entries of lower level; otherwise leave σ_w free. Then the free entries at level n are exactly those indexed by the Lyndon words of length n . This is the elimination of Section 3 of the main text.*

Proof. Let w be non-Lyndon with Lyndon factorisation $w = l_1 \cdots l_k$, $k \geq 2$, so every factor has length below n . Evaluating the triangular identity (S3.1) on a path — legitimate because $w \mapsto \sigma_w$ is a character of the shuffle algebra, and the k -fold shuffle is generated by the pairwise identities (4) — gives

$$n_1! \cdots n_r! \sigma_w = \sigma_{l_1} \cdots \sigma_{l_k} - \sum_{v < w} c_v \sigma_v,$$

a relation of the required form: the product on the right involves lower levels only, and every v in the sum is a smaller word of length n , already processed. So every non-Lyndon word is solved.

Let w be Lyndon and suppose, for contradiction, that some relation expresses σ_w as a polynomial Q in smaller words of length n and lower-level entries. Replace each smaller non-Lyndon word by its polynomial P_v (Lemma 5); Q becomes a polynomial in Lyndon entries other than σ_w , and $\sigma_w - Q = 0$ holds on every group-like element. That is a non-trivial polynomial relation among the Lyndon entries of length $\leq n$, which Lemma 5 rules out. So every Lyndon word stays free. The count checks: the free set has $\ell_d(n)$ elements, the dimension of the level- n solution space by Lemma 5, so no relation is left unused.

The rule of thumb used in the computation follows: within a rotation class of length- n words the free word is the lexicographically smallest rotation, which by Lemma 4 is the Lyndon word of the class when the class is primitive; a non-primitive class u^k has no Lyndon member and is eliminated entirely, each of its words by the triangular identity for its own Lyndon factorisation (for u Lyndon, that of u^k is u repeated k times, and the identity reads $\sigma_u^k = k! \sigma_{u^k} + (\text{smaller words})$). $\square$

Proof of Proposition 1 of the main text. Lemma 6 identifies the core set C_m produced by the elimination with the Lyndon words of length $\leq m$, which is the first statement. Lemma 5 identifies the corresponding entries with a transcendence basis of the coordinate ring of $G^m(\mathbb{R}^d)$, which is the second. The substitution rules Q_m are the polynomials P_v of Lemma 5, written out for each non-Lyndon v with $|v| \leq m$ (Section S1 of the supplement); they are the inverse of the isomorphism (S3.2) on the chosen representatives, which is the third. $\square$

The count $|C_m| = \sum_{n \leq m} \ell_d(n)$ and the Pascal-triangle refinement of Section 3 of the main text follow from Theorem 1 and the multi-graded Witt formula.

S4. Proofs of the stability theorems of the main text (Theorems 7, 9 and 14)

Proof of Theorem 7 of the main text. Write $f(t) = \sum_{i=1}^D \alpha_i t^i$. By Proposition 5 of the main text, for $l = 1, \dots, D$,

$$\sigma_{(1^{l-1}, 2)}[X_f] = \sum_{i=1}^D \alpha_i \frac{i T^{l+i-1}}{(l-1)!(l+i-1)} =: (M\alpha)_l,$$

and M is the scaled Hilbert matrix of Corollary 6 of the main text with the rows multiplied by $T^{l-1}/(l-1)!$ and the columns by T^i , hence invertible. These D coordinates are linear in α , so $E_D(f, g) \geq q_{\min} \|M(\alpha - \alpha')\|^2 \geq q_{\min} s_{\min}(M)^2 \|\alpha - \alpha'\|^2$ with $s_{\min}(M) > 0$ the smallest singular value, for all f, g . Every other retained coordinate is a polynomial in α (the entries of

a polynomial path are polynomials in its coefficients, by (2)), so on a closed ball containing K its gradient is bounded and the mean-value inequality gives $E_D(f, g) \leq C' \|\alpha - \alpha'\|^2$ with C' depending on the ball. Finally $\mathcal{P}_D^0$ is finite-dimensional, so the coefficient norm and the $H^s(0, T)$ norm are equivalent on it, with constants depending on D , T and s . $\square$

Proof of Theorem 9 of the main text. Choose an invertible p -row minor F of $DS_m(\theta_0)$ and apply the inverse function theorem. For any further coordinate h , the function $h \circ F^{-1}$ is C^1 , proving the factorisation assertion. On a sufficiently small convex ball about θ_0 , continuity of DF makes $\sup \|DF - DF(\theta_0)\|$ smaller than half the smallest singular value of $DF(\theta_0)$. Integrating derivatives along segments then bounds $|F(\theta) - F(\eta)|$ below by a positive constant times $|\theta - \eta|$. A bound on DS_m gives the upper estimate. If the path derivative into H^s is injective, its minimum norm on the unit sphere of $\mathbb{R}^p$ is positive. The same derivative argument in that Hilbert space gives local upper and lower bounds for $\|f_\theta - f_\eta\|_{H^s}$ in parameter distance. Combining the estimates proves the last assertion. $\square$

Proof of Theorem 14 of the main text. Lipschitz functions are absolutely continuous with $|f'| \leq L$ almost everywhere. For a word of length k , simplex integration gives $|\sigma_w[(t, f)]| \leq A^k T^k / k!$. There are at most 2^k words at that level, so the total weighted squared difference there is at most $4(2A^2 \rho^2)^k$. The resulting geometric majorant proves uniform convergence. The distance is the norm of the difference of two weighted core vectors in ℓ^2 , hence satisfies the triangle inequality.

If $d_\rho(f, g) = 0$, the level-one value coordinate gives $f(T) = g(T)$. Proposition 5 of the main text then shows that all ordinary moments of $f - g$ vanish. Density of polynomials in $C([0, T])$ implies $f = g$, establishing positive definiteness.

Each fixed iterated integral is continuous for uniform convergence on K_L . To see this directly, let $X_n \rightarrow X$ uniformly with a common variation bound, and induct on the length of the word using the running integrals $I_{wi}(t) = \int_0^t I_w dX^i$. The integral of $(I_w^n - I_w)$ against dX_n^i tends to zero by the variation bound. For the integral of I_w against $d(X_n^i - X^i)$, integration by parts bounds it by a constant times $\|X_n - X\|_\infty$, since the fixed I_w has bounded variation. This proves the induction. Uniform convergence of the series now makes $f \mapsto (\sqrt{W_{|w|}} \sigma_w[(t, f)])_w$ continuous into ℓ^2 .

The class K_L is compact in the uniform topology by Arzelà–Ascoli. A continuous injection of a compact space into a metric space is a homeomorphism onto its image, and its inverse is uniformly continuous. Explicitly, set $\omega(r) = \sup\{\|f - g\|_\infty : f, g \in K_L, d_\rho(f, g) \leq r\}$. Compactness and injectivity imply $\omega(r) \rightarrow 0$, as claimed. $\square$